1. Microsoft Excel 2007 – Overview

Excel 2007 is the latest and greatest spreadsheet application from Microsoft. It is one of the many software programs that make up MS Office 2007 which is available only for the Windows PC operating system. Excel is a program which displays a table of numbers and text in rows and columns, and used for accounting, budgeting, financial analysis, scientific applications, and other work with figures.

2. Introduction to MS excel sheet

a. **Title bar:** File name, format, mode, application.
b. **Quick Access Toolbar:** Save, Undo, redo etc.
c. **The Office button:** New, Open, Save, and so on.
d. **The Excel Ribbon:** Home, Insert, Page Layout, and so on.

3. Creating and saving workbooks

a. When opening Excel 2007 it automatically starts with a new **workbook.**
b. To close that workbook, select **Office Button > Close**
c. **Creating a new workbook**
   i. To create a new **workbook**, select **Office Button > New.**
   ii. Every new workbook will have three worksheets **(Sheet 1, Sheet 2, Sheet 3)** located in the bottom left corner of the Excel window. You can create more or delete the ones you already have.
d. **Saving a workbook**
   i. To save a workbook (file), select **Office Button > Save** or **Save As**
   ii. Keyboard shortcut: **Ctrl+S**

4. Entering, modifying, and deleting data

a. **Enter data**
   i. Click once (select) on cell **A1** and type. Then tap the **Enter (return)** key.
b. **Modify (edit) data**
   i. By clearing previous data first: Click once on cell **A1** and type new data.
   ii. By editing current data: Double click inside cell **A1,** with your arrow keys move your insertion point (blinking cursor) and edit data.
c. **Selecting:** Highlight or select the text using the mouse. To select the entire **Worksheet** click on the top-left of the **worksheet,** where the **column** ID's and **row** ID's intersect.

5. Printing basics

a. Always save your workbook first, **Ctrl+S.**
b. Select, **Office Button > Print > Print Preview.**
c. Click on the page to **Zoom** in or out.
d. Review the **Ribbon** as it has many options in **Print Preview.**
e. If you were satisfied with the way it would print then you would select **Print** or if you wanted to continue to edit the worksheet then you would select the **Close Print Preview** button.

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Lesson: 1 – 2 (Use of formula and function in excel)

While using function in excel sheet, one should keep the followings in mind-

- Keep the cell pointer in that cell to which the output will be displayed;
- Type = before typing the formula/function;
- To get the output, press ENTER button after typing the formula.

❖ Practice Problem- 1: Use of sum and sumif function

Mr. Afzal purchased the following items from a local market. Find the total of his expenditure.

	A	B	C
1	Item	Price	
2	Fish	120	
3	Dal	50	
4	Vegetables	56	
5	Egg	80	
6	Butter Oil	240	
7	Onion	100	
8	Ginger	80	
9	Rice	300	
10	Potato	50	
11			

Use the following formula/function.

= b2+b3+b4+b5+b6+b7+b8+b9+b10

or

=sum(b2,b3,b4,b5,b6,b7,b8,b9,b10)

or

=sum(b2:b10)

❖ **Conditional sum:** Find the total price of the items of the price greater than 100 Tk. only.

Use the following formula/function.
=sumif(b2:b10,">100")

	A	B	C	D
1	Name	Designation	Salary	
2	Dr. Abdus Salam	Professor	24500	
3	Nazmul Hasan	Assistant Professor	13000	
4	Dr. Sukalyan Kundu	Assistant Professor		
5	Khan Asif Afzal	Lecturer	9500	
6	Fahima Tabassum	Associate Professor	14000	
7	Dr. Rassel Akram	Professor	26000	
8	Khan Ataus Samad	Associate Professor	15000	
9	Jesmin Akhter	Lecturer	7200	
10	Atiqul Islam	Assistant Professor	12000	
11	Mahmooda Ghani	Professor	27000	
12				
13		**Professor**		
14		**Associate Professor**		
15		**Assistant Professor**		
16		**Lecturer**		
17				

To find out the total salary of professor only type

= sumif(b2:b11,"professor",c2:c11)

Similarly, the total salary of assistant professor, associate professor and lecturer can be obtained.

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❖ **Practice Problem- 2: Subtraction**

Find out the total profit or loss based on the following table.

	A	B	C	D	E
1	**Item**	**Buying Price**	**Selling Price**	**Difference**	
2	T-Shirt	1200	1350		
3	Jeans	1500	1800		
4	Panjabi	700	640		
5	Sari	2000	2500		
6	Lungi	350	250		
7	Fotua	200	230		
8	Jacket	2100	2400		
9	Blazer	4000	4500		
10					
11		**Total Profit or Loss**			
12					

Functions

i) Keep the cell pointer at **d2** cell and type **=c2-b2** and then press **ENTER**.

ii) Keep the cell pointer at **d2** and drag the fill handle up to **d9**.

iii) Keep the cell pointer at **d11** and type **=sum(d2:d9)** and then press **ENTER**.

❖ **Practice Problem- 3: Multiplication**

Find out the total grand total of the prices based on the following table.

	A	B	C	D	E
1	**Items**	**Quantity**	**Unit Price**	**Total Price**	
2	Fish	5	120		
3	Dal	2	100		
4	Vegetables	1	25		
5	Egg	20	6		
6	Butter Oil	3	60		
7	Onion	5	25		
8	Ginger	1	60		
9	Rice	20	30		
10	Potato	7	12		
11					
12		**Grand Total**			
13					

Functions

i) Keep the cell pointer at **d2** cell and type **=product(b2,c2)** and then press **ENTER**.

ii) Keep the cell pointer at **d2** and drag the fill handle up to **d10**.

iii) Keep the cell pointer at **d12** and type **=sum(d2:d10)** and then press **ENTER**.

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❖ Practice Problem- 4: Division

Find out the unit price for each items based on the following table.

	A	B	C	D	E
1	Item	Quantity	Total Price	Unit Price	
2	Fish	5	600		
3	Dal	2	200		
4	Vegetables	1	25		
5	Egg	20	120		
6	Butter Oil	3	240		
7	Onion	5	125		
8	Ginger	1	60		
9	Rice	20	600		
10	Potato	7	84		
11					

Functions

i) Keep the cell pointer at **d2** cell and type **=c2/b2** and then press **ENTER.**

ii) Keep the cell pointer at **d2** and drag the fill handle up to **d10.**

❖ **Practice Problem- 5: Average**

Find out the average temperature of each cities and hence of the country based on the following table.

	A	B	C	D	E	F
1	City	Temp. at Morning	Temp. at Noon	Temp. at Evening	Temp. at Night	Average Temp. of each City
2	Dhaka	25	31	28	22	
3	Rajshahi	27	30	29	24	
4	Khulna	23	29	25	25	
5	Chittagong	25	31	31	23	
6	Sylhet	21	27	26	20	
7	Barisal	26	30	29	24	
8						
9		Average Temperature of the Country				
10						

Functions

i) Keep the cell pointer at **f2** cell and type **=average(b2:e2)** and then press **ENTER.**

ii) Keep the cell pointer at **f2** and drag the fill handle up to **f7.**

iii) Keep the cell pointer at **f9** and type **=average(f2:f7)** and then press **ENTER.**

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❖ **Practice Problem- 6: Maximum and minimum**

Find out the maximum and minimum temperature based on the following table.

	A	B	C	D	E	F
1	City	Temp at Morning	Temp at Noon	Temp at Evening	Temp at Night	
2	Dhaka	25	31	28	22	
3	Rajshahi	27	30	29	24	
4	Khulna	23	29	39	25	
5	Chittagong	25	31	31	23	
6	Sylhet	21	27	26	20	
7	Barisal	26	30	12	24	
8						
9			Maximum Temp			
10			Minimum Temp			
11						
12						
13						

Functions

i) Keep the cell pointer at **d9** cell and type **=max(b2:e7)** and then press **ENTER**.

ii) Keep the cell pointer at **d10** and type **=min(b2:e7)** and then press **ENTER**.

❖ **Practice Problem- 7: Preparing Salary sheet**

Prepare a salary sheet using the following table.

	A	B	C	D	E	F	G	H	I
1	Name	Age	Basic Salary	House Rent	Medical Allowance	Transport Allowance	Bonus	Total Salary	P.fund
2	Asif	18	15000	5000	500	300			
3	Shilpi	26	12500	4500	450	300			
4	Rassel	20	14000	4800	500	300			
5	Halima	21	7500	3000	450	300			
6	Samad	22	12000	4000	500	300			
7	Alik	19	6000	3500	400	300			
8	Lovely	20	10000	3100	400	300			
9									
10									

Net Pay

Functions

i) Consider,

bonus = [Basic – (House rent + Madical allowance + Transport allowance)]/2

ii) Keep the cell pointer at **g2** cell and type **=(c2–sum(d2:f2))/2** , then press **ENTER**.

iii) Keep the cell pointer at **g2** and drag the fill handle up to **g8**.

iv) Keep the cell pointer at **h9** and type **=sum(c2:g2)** and then press **ENTER**.

v) Keep the cell pointer at **h2** and drag the fill handle up to **h8**.

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❖ **Practice Problem- 8: Use of IF function**

Comment whether pass or fail based on the following.

	A	B	C	D	E	F
1	**Name**	**Roll**	**Sex**	**Total Marks**	**Pass/Fail**	**Bonus Marks**
2	Asif	1	Male	703		
3	Moon	2	Female	662		
4	Rassel	3	Male	546		
5	Atik	4	Male	670		
6	Shilpi	5	Female	435		
7	Samad	6	Male	701		
8	Reena	7	Female	600		
9	Saddam	8	Male	644		
10	Sagar	9	Male	701		
11	Halima	10	Female	560		
12	Ranju	11	Male	872		
13	Nazu	12	Male	450		
14	Liton	13	Male	600		
15	Babu	14	Male	592		
16						

Functions

i) Consider, total 600 is the pass marks.

ii) Keep the cell pointer at **e2** cell and type **=if(d2>=600,"Pass","Fail")** , then press **ENTER**.

iii) Keep the cell pointer at **e2** and drag the fill handle up to **e15**.

❖ **Practice Problem- 9: Preparation of marksheet**

Prepare a grade sheet by filling up the following table.

	A	B	C	D	E	F	G	H
1	12	Course No.	Course Title	Credit Hour	Marks obtained	Subject based grade point	GPA	Obtained letter Grade
2	1	Chem. 110	Physical Chemistry-I	F	76			
3	2	Chem. 120	Inorganic Chemistry-I	F	82			
4	3	Chem. 130	Organic Chemistry- I	F	89			
5	4	Chem. 151	Calculus & Analytical Geometry	F	65			
6	5	Chem. 161	Properties of Matter and wave	F	76			
7	6	Chem. 162	Electricity and Magnetism	F	63			
8	7	Chem. 115	General Chemistry Practical	F	72			
9	8	Chem. 125	Qualitative Inorganic Analysis	F	86			
10	9	Chem. 150	Viva voce	F	82			
11								

Functions

i) Keep the cell pointer at **f2** cell and type-

=if(e2>=80,4,if(e2>=75,3.75,if(e2>=70,3.5,if(e2>=65,3.25,if(e2>=60,3,if(e2>=55,2.75,if(e2>=50,2.5,if(e2>=45,2,if(e2>=40,1,0))))))))) , then press **ENTER**.

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ii) Keep the cell pointer at f2 and drag the fill handle up to f10.

iii) Keep the cell pointer at **g5** and type **=average(f2:f10)** , and press ENTER.

iv) Keep the cell pointer at **h5** cell and type-

=if(g5>=80,"A+",if(g5>=75,"A",if(g5>=70,"A–

",if(g5>=65,"B+",if(g5>=60,"B",if(g5>=55,"B–

",if(g5>=50,"C+",if(g5>=45,"C",if(g5>=40,"D","F")))))))) , then press ENTER.

Exam V.V.I

Practice Problem- 9: Graph plotting

Draw volume versus temperature graph and find the slope and intercept.

	A	B
	Temperature	Volume
1	(°C)	(cm^3)
2	-205	6.7
3	-183	8.9
4	-162	10.6
5	-53	19.4
6	-23	21.8
7	-3	23.4
8	15	25.1
9	35	26.8
10	45	28.1
11	56	29.1
12		

- Click on **insert**;
- Click **scatter**;
- Select the scattered points;
- Click on the right button of the mouse keeping the cursor on the rectangular chart as appeared;
- Click on **select data**;
- Click on **add**;
- Write the series name;
- Select series x-values by dragging mouse;
- Select series y-values similarly;
- Click on **OK** and again **OK**;
- Click on the right-down icon of the chart layout tools;
- Select the layout having function (**layout 9**);
- Drag the equation box aside;
- Change the axis title and finally **save** it.

Shonta

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